Algebra: Simplifying Rational Expressions

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Name: ______________________ Date: ___________ Score: ___ / 10

DIRECTIONS

Simplify each rational expression completely. Factor where needed and cancel common factors. State any restrictions on x.

1. Simplify; state restrictions: $\dfrac{6x^2}{3x}$ Answer: ______________________	**2.** Simplify; state restrictions: $\dfrac{x^2+7x+12}{x+4}$ Answer: ______________________
3. Simplify; state restrictions: $\dfrac{4x^2-1}{2x+1}$ Answer: ______________________	**4.** Simplify; state restrictions: $\dfrac{x^3-x}{x^2-1}$ Answer: ______________________
5. Simplify; state restrictions: $\dfrac{2x^2+x-3}{x-1}$ Answer: ______________________	**6.** Simplify; state restrictions: $\dfrac{x^2-4x+4}{x^2-4}$ Answer: ______________________
7. Simplify; state restrictions: $\dfrac{3x^2-12}{x-2}$ Answer: ______________________	**8.** Simplify; state restrictions: $\dfrac{x^2+6x+9}{x+3}$ Answer: ______________________
9. Simplify; state restrictions: $\dfrac{x^2-7x+10}{x^2-4x-5}$ Answer: ______________________	**10.** Simplify; state restrictions: $\dfrac{5x^2+10x}{x+2}$ Answer: ______________________

Based on the Numberbender lesson "ALGEBRA: Simplifying Rational Expressions" • youtu.be/MltJKMq_V4g

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Answer Key & Solutions

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TEACHER NOTES

Items 1–2: factor out the GCF. Items 3–4: apply difference of squares or factor a common variable. Items 5–10: factor both numerator and denominator as trinomials or perfect squares, then cancel.

1. Simplify; state restrictions: **$6x^2 / (3x)$**

Answer: $2x$, $x \neq 0$

$6x^2/(3x) = 2x$. Restriction: $x \neq 0$.

2. Simplify; state restrictions: **$(x^2 + 7x + 12) / (x + 4)$**

Answer: $x + 3$, $x \neq -4$

Factor: $(x+4)(x+3)/(x+4)$. Cancel $(x+4) \rightarrow x+3$.

3. Simplify; state restrictions: **$(4x^2 - 1) / (2x + 1)$**

Answer: $2x - 1$, $x \neq -\frac{1}{2}$

Difference of squares: $(2x-1)(2x+1)/(2x+1)$. Cancel $\rightarrow 2x-1$.

4. Simplify; state restrictions: **$(x^3 - x) / (x^2 - 1)$**

Answer: x, $x \neq \pm 1$

Factor: $x(x^2-1)/(x^2-1)$. Cancel $(x^2-1) \rightarrow x$. Restrictions: $x \neq \pm 1$.

5. Simplify; state restrictions: **$(2x^2 + x - 3) / (x - 1)$**

Answer: $2x + 3$, $x \neq 1$

Factor: $(2x+3)(x-1)/(x-1)$. Cancel $(x-1) \rightarrow 2x+3$.

6. Simplify; state restrictions: **$(x^2 - 4x + 4) / (x^2 - 4)$**

Answer: $(x - 2) / (x + 2)$, $x \neq \pm 2$

Num: $(x-2)^2$. Den: $(x+2)(x-2)$. Cancel one $(x-2)$.

7. Simplify; state restrictions: **$(3x^2 - 12) / (x - 2)$**

Answer: $3(x + 2)$, $x \neq 2$

Factor: $3(x+2)(x-2)/(x-2)$. Cancel $(x-2) \rightarrow 3(x+2)$.

8. Simplify; state restrictions: **$(x^2 + 6x + 9) / (x + 3)$**

Answer: $x + 3$, $x \neq -3$

Num: $(x+3)^2$. Cancel one $(x+3) \rightarrow x+3$. Restriction: $x \neq -3$.

9. Simplify; state restrictions: **$(x^2 - 7x + 10) / (x^2 - 4x - 5)$**

Answer: $(x - 2) / (x + 1)$, $x \neq 5, -1$

Num: $(x-5)(x-2)$. Den: $(x-5)(x+1)$. Cancel $(x-5)$.

10. Simplify; state restrictions: **$(5x^2 + 10x) / (x + 2)$**

Answer: $5x$, $x \neq -2$

Factor GCF: $5x(x+2)/(x+2)$. Cancel $(x+2) \rightarrow 5x$.

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